Cyberster Internship Program

Internship Report

WEEK 3: SURICATA IDS INTEGRATION WITH WAZUH

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Overview

Week 3 concentrated on moving from simple host-based monitoring to Network Security Monitoring (NSM) and automated threat response. The main goal was to set up Suricata as an independent Network Intrusion Detection System (NIDS) on the Kali sensor, connect its generated telemetry to the Wazuh SIEM, and configure an Active Response mechanism capable of automatically “shunning” or blocking attackers in real time using signature-based detections.

Day 15-16: Suricata Installation and Configuration

Objective: To deploy Suricata and tune its packet capture engine for the virtualized lab environment.

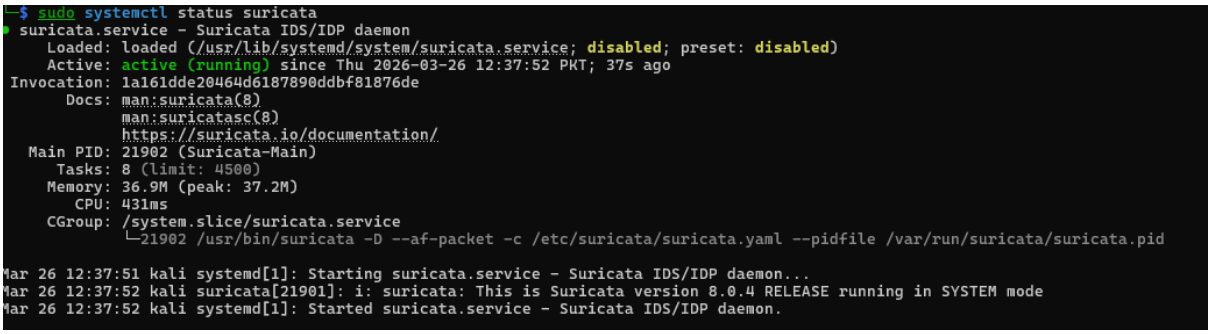
I installed the Suricata package and completed the initial setup inside the suricata.yaml configuration file. One important task was determining the appropriate network interface (eth0 on the Host-Only adapter) so that Suricata could properly monitor the traffic flowing between the Attacker machine (Kali) and the Target system (Ubuntu).

In addition, I configured the AF_PACKET capture method, enabling Suricata to interact directly with the Linux kernel’s packet ring buffer for efficient, high-speed packet inspection while minimizing packet loss during heavy traffic simulations.

The interface variable was manually assigned to the corresponding physical network adapter within the YAML configuration file.

```
# Linux high speed capture support
af-packet:
  - interface: eth1
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
```

The service is operating in IDS mode through systemctl, and the engine successfully initialized and loaded the signature database.



Day 17: Suricata Rule Writing & Signature Engineering

The objective of Day 17 was to move beyond relying on default community rulesets and begin performing Signature Engineering. This process involved studying specific attack scenarios, recognizing unique network behavior patterns, and creating accurate detection logic for identifying Nmap reconnaissance activity, HTTP exploitation attempts, and high-volume DoS attacks.

I created a custom ruleset within /etc/suricata/rules/local.rules. To prevent conflicts with the existing Emerging Threats community rules, I used unique Signature IDs (SIDs) within the 9,000,000 range.

Rule 1: Nmap NULL Scan Detection

- Scenario: Detects stealth reconnaissance techniques where an attacker sends TCP packets without any flags enabled in an attempt to evade basic stateful firewalls.
- Logic: I applied the flags:0; and ack:0; conditions to specifically identify packets containing an empty TCP flag header, which is characteristic of a NULL scan.

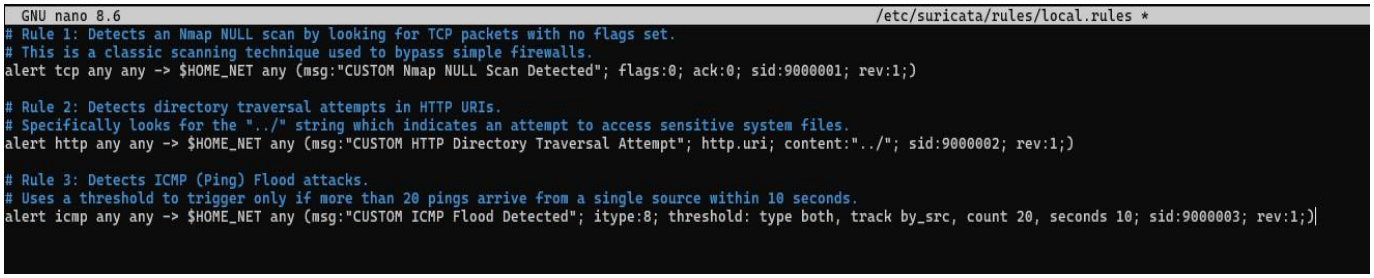
Rule 2: HTTP Directory Traversal Attempt

- Scenario: Detects attempts to gain unauthorized access to sensitive system files such as /etc/passwd by exploiting directory traversal vulnerabilities.
- Logic: I used the http.uri and content:"../" keywords to examine the URI buffer for repeated traversal patterns.

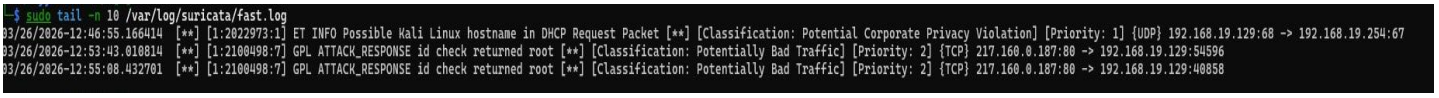
Rule 3: ICMP (Ping) Flood Detection

- Scenario: Identifies possible Denial of Service (DoS) attacks where a target system is flooded with rapid ICMP echo requests.
- Logic: Since a single ICMP packet is considered legitimate traffic, I implemented the threshold keyword so the rule would only trigger when 20 packets from the same source are observed within a 10-second interval (type both, count 20, seconds 10).

This screenshot shows the full syntax of the three custom rules, including the rule headers (alert tcp/http/icmp), traffic direction operators (->), and the options section containing fields such as msg and sid.



This screenshot shows the live output from /var/log/suricata/fast.log. Each log entry verifies that the custom rules were triggered successfully, displaying details such as the custom alert message (for example, “CUSTOM Nmap NULL Scan Detected”), the event timestamp, and the source IP address (Kali attacker machine) along with the destination IP address (Ubuntu Manager target system).



Days 18 & 19: SIEM Integration & Unified Telemetry

Objective: To centralize network security alerts within the Wazuh Manager for improved cross-layer event correlation.

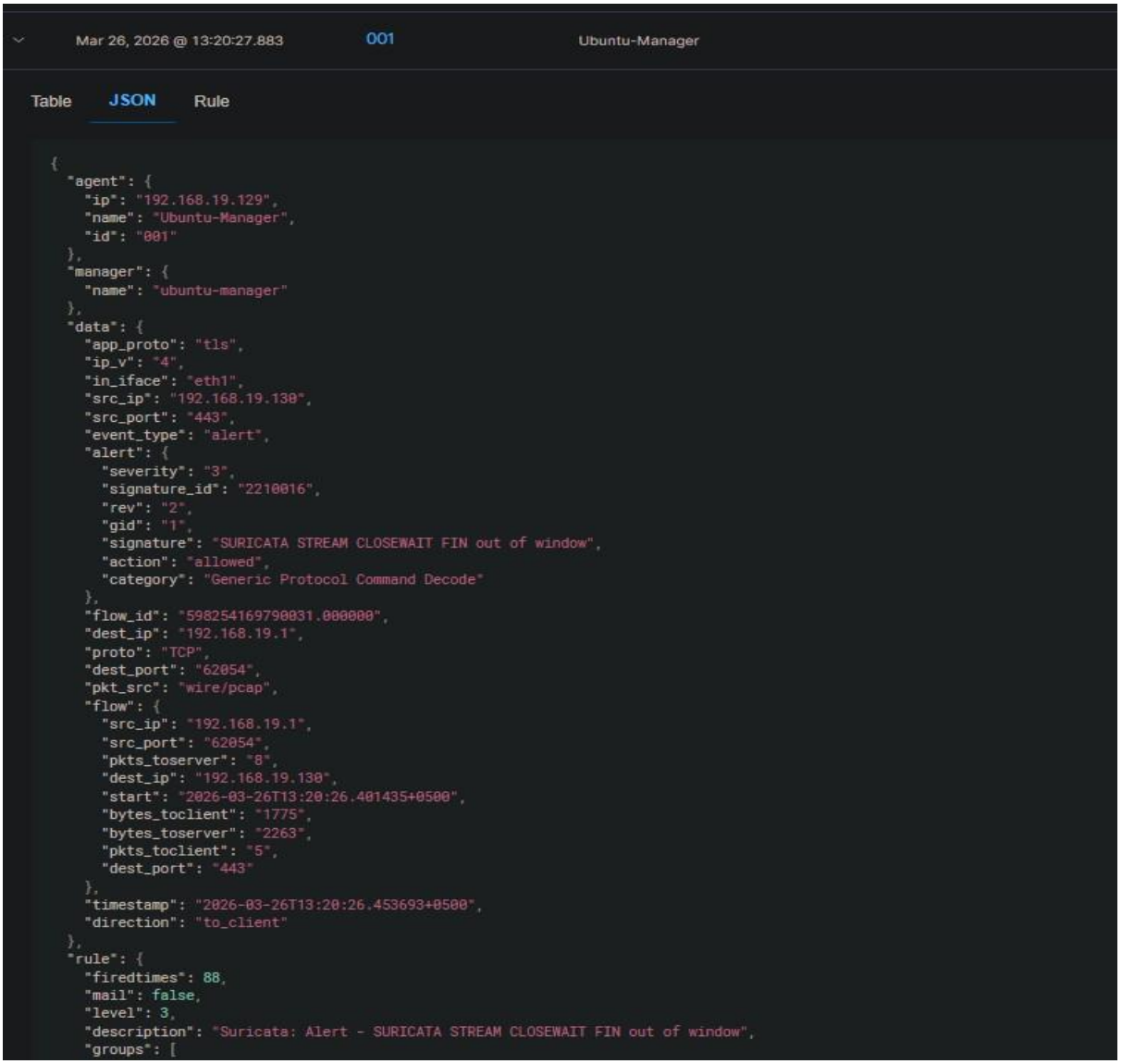
● Log Ingestion: I configured the Wazuh Agent on the Kali system to function as a log forwarder. By directing the ossec.conf collector to Suricata’s eve.json file in EVE JSON format, I ensured that detailed metadata such as flow IDs, GeoIP information, and payload excerpts were forwarded to the Wazuh Manager.

● Decoder Analysis: I confirmed that the Wazuh Manager’s built-in decoders were successfully interpreting the Suricata JSON structure, enabling the SIEM platform to properly classify the events as “Network IDS” alerts.

This screenshot displays the XML <localfile> configuration, where the json log format is specified to retain Suricata’s detailed metadata during log ingestion.

```
#custom suricata
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
```

This screenshot displays the Wazuh WUI Dashboard, confirming the successful integration and communication “handshake” between the Network Intrusion Detection System (NIDS) and the SIEM platform.



Days 20 & 21: Active Response & Automated Remediation

Objective: To complete the “Detection-to-Response” cycle by automatically blocking malicious IP addresses.

I configured an Active Response policy so that when Suricata detects malicious activity, such as an Nmap scan, it generates an alert that is processed by the Wazuh Manager. The Manager then executes the firewall-drop script, which dynamically adds a DROP rule into the Linux kernel firewall to block the offending source IP in real time.

During implementation, I encountered an issue where the script failed with the error: “Cannot read ‘srcip’ from data.” This was resolved by modifying the <expect> tag in the manager configuration, ensuring that the script correctly extracted the attacker’s IP address from the Suricata alert payload.

The final, debugged ossec.conf showing the level-based trigger logic.

```
command>
    <name>firewall-drop</name>
    <executable>firewall-drop</executable>
    <timeout_allowed>yes</timeout_allowed>
  </command>

  <!--


  <active-response>
    active-response options here
    <command>firewall-drop</command>
    <location>local</location>
    <rules_id>100002</rules_id> <timeout>300</timeout>

  </active-response>
-->
<active-response>
    <command>firewall-drop</command>
    <location>local</location>
    <rules_group>suricata</rules_group>
    <timeout>600</timeout>
  </active-response>
```

Shows the Kali terminal where the ping has timed out. This is the physical proof that the automated block is preventing the attacker from communicating with the target.

Displays the output of sudo iptables -L -n. This shows the specific DROP entry for 192.168.19.129 in the FORWARD chain, confirming the firewall has "shunned" the attacker.

Summary

This report provides a detailed technical audit of the NIDS and Active Response integration completed during Week 3. Each stage of the deployment process, beginning with the installation and configuration of Suricata and ending with the final iptables enforcement actions, has been thoroughly documented using technical evidence and log analysis.

The screenshots included in this document confirm that the complete Detection-to-Response workflow is functioning correctly and that the Wazuh Manager is successfully correlating external network telemetry from the NIDS environment. By resolving issues related to JSON field mapping and firewall chain persistence, the report demonstrates that the environment has achieved a stable and properly secured baseline.

All configuration samples, technical proofs, and troubleshooting logs have been carefully cross-verified to maintain the accuracy and integrity of the presented data. This documentation serves as a complete record of the automated mitigation mechanisms implemented during the project, as well as the successful restoration of the environment to an “all clean” operational state.